

LAKE MURRAY, SC, USA

WMO: 998177

Lat: 34.1074N

Lon: 81.2707W

Elev: 109

StdP: 100.02

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.5	(c) -1.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 11.2	(k) 11.6	(l) 10.4	(m) 12.8	(n) 1.9	(o) 20	(p) 0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.5	(c) 34.7	(d) N/A	(e) 33.6	(f) N/A	(g) 32.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.3	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.7	(b) 7.5	(c) 6.4	(e) -5.8	(f) 37.7	(g) 3.1	(h) 1.8	(i) -8.0	(j) 39.0	(k) -9.8	(l) 40.0	(m) -11.5	(n) 41.0	(o) -13.8	(p) 42.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	7.8	9.9	12.9	18.4	23.2	27.1	28.1	27.3	24.7	18.9	12.8	10.5		
	DBStd	8.16	5.13	4.90	4.65	3.84	3.41	2.35	1.89	2.21	3.09	3.92	4.10	4.77		
	HDD10.0	263	106	58	24	1	0	0	0	0	0	1	20	53		
	HDD18.3	1249	328	239	176	46	7	0	0	0	1	40	168	245		
	CDD10.0	3372	36	54	115	254	410	513	562	536	442	277	104	69		
	CDD18.3	1316	0	2	8	49	158	263	303	278	193	58	2	2		
	CDH23.3	11650	0	6	32	226	1074	2539	3212	2741	1538	275	6	1		
	CDH26.7	4370	0	1	2	38	328	1022	1331	1078	514	56	1	0		
(14) Wind	WSAvg	2.4	2.6	2.6	2.6	2.7	2.6	2.6	2.7	2.0	1.8	2.1	1.9	2.3		
(15) Precipitation	PrecAvg	1158	101	95	111	72	74	123	112	115	96	89	78	90		
	PrecMax	1441	216	185	244	180	146	239	230	202	226	283	181	227		
	PrecMin	774	41	49	27	16	20	35	44	30	5	0	25	31		
	PrecStd	188	46	36	50	43	32	61	53	50	57	63	44	49		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.6	23.8	26.6	30.2	34.1	36.2	35.8	35.7	33.9	31.3	23.7	22.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	18.6	20.8	23.9	27.5	31.5	34.3	34.7	34.0	32.4	28.0	21.8	20.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	16.9	18.8	21.8	25.9	30.0	32.8	33.4	32.8	31.1	26.0	20.5	19.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.2	9.8	9.8	9.1	8.9	8.5	8.6	8.5	9.3	9.6	8.3		
		MCDBR	10.0	10.9	12.1	11.5	10.7	10.4	9.9	10.1	10.2	10.6	11.4	8.7		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.330	0.369	0.410	0.473	0.503	0.550	0.526	0.444	0.399	0.350	0.321		
	taud		2.494	2.467	2.388	2.263	2.136	2.094	1.984	2.039	2.254	2.348	2.444	2.530		
	Ebn at Noon		890	919	905	878	823	795	755	766	818	829	845	869		
	Edh at Noon		85	97	114	135	155	162	179	167	129	108	89	78		
(44) All-Sky Solar Radiation	RadAvg		2.75	3.38	4.46	5.61	6.13	6.41	6.06	5.53	4.76	3.97	3.03	2.39		
	RadStd		0.14	0.36	0.33	0.42	0.48	0.45	0.40	0.27	0.41	0.48	0.24	0.23		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+176	+121

Nomenclature: See separate page